

Pulsed NMR Relaxometry of Beeswax Across the Solid-to-Liquid Phase Transition

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1 Theory

1.1 Nuclear Magnetic Resonance

Atomic nuclei with non-zero spin carry a magnetic moment $\boldsymbol{\mu} = \gamma\hbar\mathbf{I}$, where γ is the gyromagnetic ratio and $\mathbf{I}$ is the dimensionless spin operator. In an external static field $\mathbf{B}_0 = B_0\hat{z}$ the energy eigenstates split by $\Delta E = \hbar\gamma B_0$, and at thermal equilibrium a small net magnetisation $M_0 \propto B_0/T$ builds up along $\hat{z}$. The ensemble magnetisation $\mathbf{M}$ precesses about $\hat{z}$ at the *Larmor frequency*

$$\omega_L = \gamma B_0. \quad (1)$$

For protons (^1H), $\gamma/2\pi \approx 42.6 \text{ MHz T}^{-1}$; at the lab field of $B_0 \approx 0.177 \text{ T}$ this gives $\omega_L/2\pi \approx 7.5 \text{ MHz}$.

Applying a transverse radiofrequency (RF) field oscillating at $\omega \approx \omega_L$ for a short pulse of duration t_p rotates $\mathbf{M}$ by angle $\alpha = \gamma B_1 t_p$ away from $\hat{z}$. A 90° pulse ($\alpha = \pi/2$) rotates $\mathbf{M}$ entirely into the transverse plane, where it precesses and induces an oscillating EMF in the pick-up coil. This voltage decays as the spins dephase — the *free induction decay* (FID).

1.2 Relaxation

Two independent mechanisms restore the system to thermal equilibrium after a pulse.

Spin–lattice relaxation (T_1). The longitudinal component recovers as

$$M_z(t) = M_0 \left(1 - e^{-t/T_1}\right), \quad (2)$$

driven by fluctuating magnetic fields from molecular motion that allow spins to exchange energy

with the lattice. T_1 sets the minimum wait time between successive pulse sequences; if the repeat time $T_R \ll T_1$ the magnetisation is partially saturated and the apparent signal amplitude is suppressed by a factor $(1 - e^{-T_R/T_1})$.

Spin–spin relaxation (T_2). The transverse component decays as

$$M_{xy}(t) = M_0 e^{-t/T_2}, \quad (3)$$

caused by randomly fluctuating local dipole fields from neighbouring nuclei inducing small shifts in each spin’s precession frequency. These random frequency offsets cause the spins to fan out in the transverse plane, irreversibly destroying the phase coherence of the ensemble. The observed FID envelope decays with the shorter time constant T_2^* defined by $1/T_2^* = 1/T_2 + \gamma\Delta B_0$, where ΔB_0 is the static field inhomogeneity across the sample.

1.3 The Spin-Echo Sequence

The T_2^* contribution from static field inhomogeneity is reversible and can be removed with the *Hahn spin-echo* sequence:

$$90_x^\circ - \tau - 180_x^\circ - \tau - \text{echo}.$$

The 90° pulse creates transverse magnetisation; during τ the spins precess at slightly different rates and dephase. The 180° pulse mirrors all spin phases about the x' -axis, reversing their relative ordering. After a further τ the spins rephase coherently, producing a signal maximum (the *echo*) at time 2τ .

Because the 180° pulse also reverses the dephasing from static inhomogeneity, the echo amplitude is insensitive to ΔB_0 and reflects only

true spin–spin dephasing and diffusion:

$$E(2\tau) = E_0 \exp\left(-\frac{2\tau}{T_2} - \frac{2}{3}\gamma^2 G^2 D \tau^3\right), \quad (4)$$

where G is the field gradient and D is the translational diffusion constant. For small τ the cubic diffusion term is negligible and $E \propto e^{-2\tau/T_2}$, providing a clean measurement of T_2 .

1.4 Beeswax as an NMR Sample

Beeswax is a complex organic mixture of long-chain esters, hydrocarbons, and fatty acids. Its proton NMR signal is dominated by the dense network of CH_2 and CH_3 groups along the carbon chains. The material passes through three thermal phases:

- **Solid (crystalline):** $T \lesssim 40^\circ\text{C}$. Chains are packed into ordered lamellae; rigid lattice severely restricts molecular motion.
- **Semicrystalline:** $40^\circ\text{C} \lesssim T \lesssim 62^\circ\text{C}$. Crystalline and amorphous domains coexist.
- **Liquid:** $T \gtrsim 62^\circ\text{C}$. Long-range order is lost; rapid isotropic tumbling averages many dipolar interactions.

In the fast-motion (extreme-narrowing) limit, rapid molecular tumbling averages the static dipole–dipole coupling and drives $T_2 \rightarrow T_1$. As temperature increases and molecular dynamics speed up, one therefore expects T_2 to lengthen. In the rigid-lattice limit, by contrast, static dipolar couplings are strong and T_2 can be extremely short ($\sim 10\ \mu\text{s}$). Beeswax at our temperatures ($45\text{--}70^\circ\text{C}$) straddles these regimes, making it an interesting system to probe with NMR.

2 Apparatus

The NMR spectrometer is built around a permanent magnet producing $B_0 \approx 0.177\text{ T}$. An Agilent 8313A frequency synthesiser generates a CW signal near ω_L ; a double-balanced mixer (DBM) gates it into pulses under control of a custom pulse programmer. A 2 W RF power amplifier drives the probe, which consists of an LC

resonant circuit tuned and impedance-matched to $50\ \Omega$ at ω_L . Crossed-diode protection on a quarter-wavelength transmission line isolates the preamplifier (Mini-Circuits ZEL-2060A+) from the high-power transmit path; the diodes are non-conducting during transmission and present a high impedance to the preamp. After amplification, the signal passes through a bandpass filter ($\approx 7.5\text{ MHz}$ centre) and is mixed with the synthesiser reference to produce a low-frequency beat signal displayed on an oscilloscope.

The beeswax sample was held in a standard 10 mm NMR glass tube inside the RF coil and repositioned to the region of most uniform field between measurements. Temperature was monitored non-invasively with a FLIR C5 compact thermal camera aimed at the sample tube; no contact thermometer was used during acquisition so as not to disturb the field environment.

3 Experimental Procedure

The beeswax was melted on a hot plate to $\approx 70^\circ\text{C}$ and then allowed to cool naturally to $\approx 45^\circ\text{C}$ with the sample sitting in the magnet bore. At each of the five pulse spacings $\tau \in \{2, 6, 10, 16, 18\}\text{ ms}$ a Hahn spin-echo sequence was run continuously while the sample cooled, and the peak-to-peak echo amplitude was recorded at $\Delta T \approx 1^\circ\text{C}$ intervals by photographing the oscilloscope screen. Each scope display was averaged over 16 successive shots to reduce noise before being captured.

The 90° and 180° pulse widths were determined at the start of each session by finding the pulse width that minimised the FID amplitude (giving t_{180}) and halving it. Values of $t_{90} = 31 \pm 2\ \mu\text{s}$ and $t_{180} = 62 \pm 5\ \mu\text{s}$ were used. The repeat time T_R was set individually for each τ channel:

τ (ms)	T_R (ms)
2	130
6	220
10	180
16	280
18	220

These repeat times were chosen to keep the echo observable on a convenient oscilloscope time-base; they were not scaled to T_1 and therefore vary across channels in a way that introduces a systematic amplitude bias (see Section ??).

4 Data Analysis

4.1 Amplitude Extraction and Pixel Calibration

Oscilloscope photographs were taken at each (τ, T) combination. The oscilloscope displayed both the spin echo waveform and a calibration bar of known amplitude, rendered by the scope's internal graticule system, at the bottom of every screen capture. Amplitudes were measured in pixels from the photographs, giving a raw peak-to-peak amplitude A_{px} with uncertainty σ_{px} estimated by the pixel-level measurement precision. Separately, the calibration bar height in pixels C_{px} (with uncertainty $\sigma_{C_{\text{px}}}$) and its known value in millivolts C_{mV} were recorded.

Because the oscilloscope vertical scale was adjusted independently for each τ channel to keep the echo on screen, a separate calibration factor

$$s_k = \frac{C_{\text{mV},k}}{C_{\text{px},k}} \quad (5)$$

was computed for each channel $k \in \{2, 6, 10, 16, 18\}$ ms. The calibration values were:

τ (ms)	C_{px}	C_{mV}
2	266	5.44
6	266	5.44
10	441	23.0
16	441	23.0
18	263	13.6

The amplitude in millivolts is then $A_{\text{mV}} = A_{\text{px}} \cdot s_k$, and the combined fractional uncertainty is

$$\frac{\sigma_A}{A} = \sqrt{\left(\frac{\sigma_{A_{\text{px}}}}{A_{\text{px}}}\right)^2 + \left(\frac{\sigma_{C_{\text{px}}}}{C_{\text{px}}}\right)^2}, \quad (6)$$

where the known calibration mV value C_{mV} is treated as exact.

4.2 Fit Region Selection

A naive application of the spin-echo decay model $A(\tau) = A_0 e^{-2\tau/T_2}$ to all five τ values fails because the repeat times T_R differ across channels. The echo amplitude from a sequence with repeat time T_R is proportional to

$$A \propto \left(1 - e^{-T_R/T_1}\right) e^{-2\tau/T_2}, \quad (7)$$

so channels with short T_R relative to T_1 are systematically suppressed. As beeswax approaches its melting point, T_1 grows large (approaching the liquid-state limit $T_1 \sim T_2 \sim 100$ ms), making the saturation factor $(1 - e^{-T_R/T_1})$ strongly channel-dependent. In particular the $\tau = 2$ ms channel ($T_R = 130$ ms) is far more saturated than the $\tau = 16$ ms channel ($T_R = 280$ ms), producing a spurious apparent rise in amplitude from short to long τ that is unrelated to T_2 .

Additionally, Eq. (4) shows that diffusion contributions grow as τ^3 ; they are negligible at large τ where the T_2 term dominates. For both reasons, fits were restricted to the three longest pulse spacings, $\tau \in \{10, 16, 18\}$ ms, which form a monotonically decaying sequence for most temperatures and sit in a regime where diffusion is small.

4.3 Exponential Decay Fitting

For each temperature T_i , the three amplitudes $\{A_{10}, A_{16}, A_{18}\}$ with uncertainties $\{\sigma_{10}, \sigma_{16}, \sigma_{18}\}$ were fit to

$$A(\tau) = A_0 e^{-2\tau/T_2} \quad (8)$$

using non-linear least squares (`scipy.optimize.curve_fit`) with `absolute_sigma=True` so that the reported covariance matrix reflects the physical uncertainties rather than the scatter. The initial guess was $A_0^{(0)} = A_{10} \exp(2\tau_{10}/15)$, $T_2^{(0)} = 15$ ms, which places the starting point in the correct order of magnitude for liquid beeswax.

With $N = 3$ data points and $p = 2$ free parameters (A_0 and T_2), the fit has $\nu = N - p = 1$ degree of freedom. The uncertainty on T_2 is taken as $\sigma_{T_2} = \sqrt{\mathcal{C}_{22}}$ where $\mathcal{C}$ is the covariance matrix returned by `curve_fit`.

4.4 Goodness-of-Fit Assessment

The chi-squared statistic is

$$\chi^2 = \sum_k \left(\frac{A_k - A(\tau_k)}{\sigma_{A_k}} \right)^2, \quad (9)$$

with the sum over the three fitted τ values. The p -value $p = P(\chi_1^2 > \chi_{\text{obs}}^2)$ is evaluated from the χ^2 distribution with $\nu = 1$ degree of freedom using `scipy.stats.chi2.sf`. A small p -value indicates that the residuals are larger than expected from the stated measurement uncertainties, signalling either underestimated errors or model inadequacy.

4.5 Temperature Uncertainty

The thermal camera records temperature at discrete frames separated by $\Delta t = 1$ s. A 1 s timing uncertainty in associating a camera frame with the moment of echo acquisition propagates to a temperature uncertainty

$$\sigma_T = \left| \frac{dT}{dt} \right| \Delta t. \quad (10)$$

The cooling rate $|dT/dt|$ varies during the run and differs slightly between τ channels (because each channel was run in a separate cooling trial). A calibration dataset (`beeswax_temp_calibration.csv`) recorded $T_{\text{cam}}(t)$ continuously during five independent cooling runs, one per τ value.

For each calibration run the derivative was estimated numerically using central finite differences at interior temperature points,

$$\left. \frac{dT}{dt} \right|_i = \frac{T_{i+1} - T_{i-1}}{t_{i+1} - t_{i-1}}, \quad (11)$$

and one-sided differences at the endpoints. The resulting $|dT/dt|(T)$ curve was interpolated to each measurement temperature T_i using linear interpolation. The per-channel estimates were then averaged to give a single uncertainty estimate at each T_i :

$$\sigma_T(T_i) = \Delta t \cdot \frac{1}{N_\tau} \sum_{k=1}^{N_\tau} \left| \frac{dT}{dt} \right|_k(T_i), \quad (12)$$

with $N_\tau = 5$.

5 Results

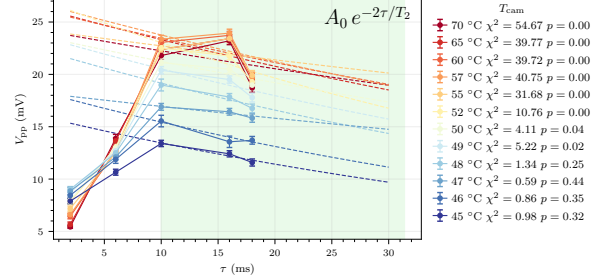


Figure 1: Peak-to-peak spin-echo amplitude vs. pulse spacing τ for beeswax at 12 temperatures (70 °C, warm colours; 45 °C, cool colours). Dashed curves are fits of Eq. (??) to the green-shaded region ($\tau \geq 10$ ms). Legend entries quote χ^2 and p -value for each fit ($\nu = 1$).

Figure ?? shows the calibrated amplitudes alongside the fitted decay curves. The data reveal two qualitatively distinct regimes.

Low temperature ($T \lesssim 49^\circ\text{C}$). The amplitudes decrease monotonically with τ across all five channels, and the exponential fits to $\tau = 10, 16, 18$ ms have $\chi^2 \lesssim 1.4$ (Table 2), consistent with the measurement uncertainties. The model describes the data well.

High temperature ($T \gtrsim 50^\circ\text{C}$). The amplitude at $\tau = 16$ ms exceeds that at $\tau = 10$ ms for most high-temperature measurements, producing a non-monotone pattern that is incompatible with pure T_2 decay. The χ^2 values increase sharply, reaching ~ 40 – 55 at 57 – 70°C ($\nu = 1$), and the p -values are effectively zero.

Table 1: Chi-squared statistics for the exponential fits ($\nu = 1$).

$T_{\text{cam}} (^{\circ}\text{C})$	χ^2	p -value
70	54.67	< 0.001
65	39.77	< 0.001
60	39.72	< 0.001
57	40.75	< 0.001
55	31.68	< 0.001
52	10.76	0.001
50	4.11	0.043
49	5.22	0.022
48	1.34	0.247
47	0.59	0.443
46	0.86	0.353
45	0.98	0.322

The origin of the high-temperature anomaly is the variation in repeat time across τ channels (Eq. (??)). As temperature rises toward and above 62°C , beeswax melts and T_1 increases substantially. With $T_1 \gg T_R$, the $(1 - e^{-T_R/T_1})$ prefactor differentially suppresses short- τ channels (small T_R) relative to long- τ channels (large T_R), causing the apparent amplitude to *increase* with τ at short spacings — exactly the pattern seen in the data.

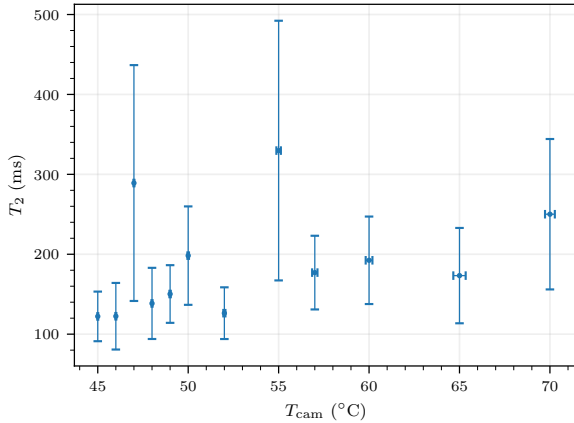


Figure 2: Fitted T_2 vs. camera temperature. Horizontal error bars are the timing-derived temperature uncertainties of Eq. (??); vertical error bars are 1σ from the covariance matrix. Only the region $T \lesssim 49^{\circ}\text{C}$ (statistically acceptable fits) is physically interpretable.

Figure ?? shows T_2 vs. T_{cam} . In the low-temperature range where fits are statistically acceptable ($T \lesssim 49^{\circ}\text{C}$), T_2 lies in the range ~ 100 – 200 ms and shows no strong trend within uncertainties. At higher temperatures the fitted T_2 values are spuriously small and scatter widely, a direct consequence of the saturation-induced amplitude bias: when the amplitude at $\tau = 16$ ms is anomalously large relative to $\tau = 18$ ms, the least-squares fit must produce a steep slope (short T_2) to accommodate both points.

The temperature uncertainty σ_T computed from Eqs. (??)–(??) is dominated by the cooling rate, which averages $\approx 0.2^{\circ}\text{C s}^{-1}$ throughout the run, giving $\sigma_T \approx 0.2^{\circ}\text{C}$. This is small compared to the 1°C spacing between measurement temperatures, so timing errors are not a dominant uncertainty source. A more significant systematic is the gap between the camera surface temperature T_{cam} and the true bulk sample temperature, which may differ by several degrees depending on the tube wall thickness and thermal equilibration.

6 Conclusion

We measured spin-echo amplitudes in beeswax at five pulse spacings across the temperature range 45 – 70°C and extracted T_2 by fitting an exponential decay model to the three longest spacings ($\tau = 10, 16, 18$ ms) at each temperature.

The goodness-of-fit analysis reveals a clear systematic failure above $\approx 50^{\circ}\text{C}$: the χ^2 values are one to two orders of magnitude above expectation, driven by a saturation bias arising from the channel-dependent repeat time T_R . In the melt phase, where T_1 grows large, the fixed T_R values suppress short- τ amplitudes relative to long- τ amplitudes, creating a spurious apparent rise in the data that the T_2 -decay model cannot fit.

In the solid/semicrystalline region ($T \lesssim 49^{\circ}\text{C}$) the fits are statistically acceptable and indicate $T_2 \sim 100$ – 200 ms. No firm conclusion about the temperature dependence of T_2 can be drawn from this limited range without either (a) scaling

T_R independently at each temperature to satisfy $T_R \gg T_1$, or (b) measuring $T_1(T)$ directly and correcting each amplitude via Eq. (??). Both improvements are straightforward with the existing apparatus and would allow the full $T_2(T)$ curve across the phase transition to be measured reliably.